

HADI ALKHUB

Brisbane, QLD | hadiaalkhub@gmail.com | github.com/HadiAlk4 | linkedin.com/in/hadiaalkhub

SUMMARY

Software engineering (Honours) student at Griffith University, focused on low-level programming, computer science theory, and sports engineering. I write systems close to the hardware — Linux drivers, bare-metal VGA, ARM64 assembly, a Git object store in Rust — and train competitive programming. Looking for a first research or engineering internship.

EDUCATION

Griffith University

Brisbane, QLD

Bachelor of Engineering, Software Engineering (Honours)

Jul. 2024 – Nov. 2027 (expected)

- GPA 6.3/7.0. Degree focus: low-level programming.
- Coursework: Advanced Computer Systems, Embedded Systems, Full Stack Development.
- Prior BIT year at Griffith (Jul. 2023 – Jun. 2024), then transferred into Software Engineering.

University of Jordan

Amman, Jordan

Coursework in German and English Literature

Oct. 2022 – Jun. 2023

- GPA 3.91/4.00. Enrolled to take selected subjects of interest (not a completed degree).

SELECTED PROJECTS

MAX7219 Device Driver & Digital Clock | *C, Linux kernel, Make*

2026

- Linux character driver (`/dev/max_screen`) for an 8-digit MAX7219 on a Raspberry Pi 4: bit-banged SPI, `ioctl` brightness, and a user-space clock with date/time, flashing seconds, and a scrolling alarm.

VGA Pong on Tiva TM4C129 | *Bare-metal C, Keil*

2026

- Software VGA on a Tiva TM4C1294NCPDT: timer HSYNC/VSYNC, GPIO RGB, a 20×30 framebuffer stretched to a full monitor (SRAM limit), ADC paddle input, and a simple opponent.

hgit | *Rust*

2026

- Git-style object database in Rust: `zlib` + SHA-1 blobs/trees/commits, with `init`, `hash-object`, `cat-file`, `write-tree`, and `commit-tree`.

Chat App System | *MEAN (MongoDB, Express, Angular, Node.js), Socket.io*

2026

- Real-time group chat: Super Admin / Group Admin / user roles, rooms, bcrypt auth, Socket.io messaging, and last-5-message retention in MongoDB.

Algorithm Visualizer | *TypeScript, React, Vite, Tailwind*

2026

- Interactive DSA visualizer (sorting, *d*-ary heaps, AVL trees, prefix/postfix evaluation).

ARM64 Assembly Calculator | *ARM64 assembly*

2026

- Command-line integer calculator in raw ARM64 assembly for Apple Silicon: ASCII conversion, `+`, `-`, `×`, `÷`, and macOS syscalls (no libc arithmetic).

COMPETITIVE PROGRAMMING

Kattis (handle `hadialk4`): ranked **24th in Australia** (score 115.9).

TECHNICAL SKILLS

Languages: C, C++, Rust, ARM64 assembly, Python, TypeScript, JavaScript, C#

Systems & tools: Linux kernel modules, Make, Git, OpenGL/GLUT, React, Vite, Tailwind CSS, Astro, Keil uVision, Raspberry Pi, Tiva TM4C129, Node.js, Express, Angular, MongoDB, Socket.io, bcrypt

Spoken: English, Arabic, German (A2)